

Why normalize $C_T = 0$?

The Eichler-integral origin of the period-polynomial normalization

Working note — conceptual scaffolding, not manuscript text

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This note exists to be read slowly and stress-tested. It answers one question: among all the representatives of the period cohomology class, why is the one with $C_T = 0$ the right one, and where did the very need to choose come from? Nothing here is meant to be pasted into the paper; it is the thing you have to believe before the paper is yours to sign.

1 The object we actually want is a primitive, not a cocycle

Fix an integer weight k , set $n := k - 2$, and let $V_n = \mathbb{C}[X, Y]_n$ carry the weight- n action of $\mathrm{SL}_2(\mathbb{Z})$,

$$(P|_\gamma)(X, Y) := P(aX + bY, cX + dY), \quad \gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix}.$$

Attach to a (weakly holomorphic, weight- k) form f the V_n -valued 1-form

$$\omega_f := f(\tau) (X - \tau Y)^n d\tau,$$

built so that ω_f is invariant under the simultaneous action on τ and on (X, Y) .

The classical period polynomial is a single integral,

$$r_f := \int_0^{i\infty} \omega_f = \int_0^{i\infty} f(\tau) (X - \tau Y)^n d\tau. \tag{1}$$

The first thing to internalize is that (1) is secretly a statement about an *antiderivative* of ω_f . Define the **Eichler integral**

$$\tilde{F}(\tau) := \int_\tau^{i\infty} f(z) (X - zY)^n dz \quad (\text{a } V_n\text{-valued function on } \mathbb{H}). \tag{2}$$

It is a genuine **primitive** (synonym: **antiderivative**) of ω_f , meaning a function whose differential returns the form: $d\tilde{F} = -\omega_f$, i.e. $\frac{d}{d\tau}\tilde{F}(\tau) = -f(\tau)(X - \tau Y)^n$. (The sign is a bookkeeping artifact of integrating *down* from $i\infty$ in (2) rather than up to τ ; orient the path the other way and it disappears. Just as a function on $\mathbb{R}$ has many antiderivatives differing by a constant, a primitive of ω_f is determined only up to an additive V_n -valued constant — the choice of fixed (upper) endpoint $i\infty$ in (2) is exactly such a choice, and that ambiguity is what §2–§6 are about.) It is transcendental, not a polynomial, and it depends on the point τ . Because the integrand transforms nicely, $\tilde{F}$ is *almost* $\mathrm{SL}_2(\mathbb{Z})$ -invariant; its failure to be invariant is the entire subject.

Claim (The cocycle is the coboundary of the primitive). *For each $\gamma \in \mathrm{SL}_2(\mathbb{Z})$,*

$$C_\gamma^f := \tilde{F} - \tilde{F}|_\gamma \iff C_\gamma^f = \int_{\gamma^{-1}} \omega_f \text{ (up to the constant } (2\pi i)^{n+1}). \quad (3)$$

This is the sentence to hang everything on. A period cocycle is not a mysterious new gadget: it is the measurement of *how badly a chosen primitive of ω_f fails to be modular*, recorded one group element at a time. The cocycle relation $C_{gh} = C_g|_h + C_h$ is then automatic — it is just

$$\tilde{F} - \tilde{F}|_{gh} = (\tilde{F} - \tilde{F}|_g)|_h + (\tilde{F} - \tilde{F}|_h)$$

rearranged — so you never have to “verify the cocycle condition” as an article of faith; it is the additivity of “ $\tilde{F}$ minus its translate.”

In one line. *A period cocycle measures the non-modularity of a primitive of ω_f . Different primitives differ by the cohomology ambiguity; the period polynomial is the non-modularity of one particular primitive.*

2 So “choose a representative” means “choose a branch of the primitive”

The cohomology class $[C^f]$ throws away the choice of primitive: adding a constant (more precisely, a V_n -valued constant of integration, or shifting the basepoint of (2)) changes $\tilde{F}$ by some fixed $P \in V_n$, hence changes every C_γ^f by the coboundary $P|_\gamma - P$. Passing to $[C^f] \in H^1(\mathrm{SL}_2(\mathbb{Z}); V_n)$ is exactly forgetting which antiderivative you used.

But a polynomial — an actual period polynomial with numbers in it — lives at the level of a *representative*, i.e. a specific primitive. So:

choosing a representative of $[C^f] =$ choosing a branch of the antiderivative $\tilde{F}$.

The question “why fix $C_T = 0$?” is therefore the question “which antiderivative is the right one to stand on?”

3 $C_T = 0$ is the cusp-regular branch

Here is the payoff. Look at what C_T measures. The slash $\tilde{F}|_T$ is the *full* weight- n action: it couples the argument shift $\tau \mapsto \tau + 1$ with the polynomial substitution $X \mapsto X + Y$ (recall $X|_T = X + Y$, $Y|_T = Y$). Evaluating the coboundary (3) as a contour gives $\tilde{F}|_\gamma(\tau) = \int_\tau^{\gamma^{-1}i\infty} \omega_f$, so for the cusp-anchored branch

$$C_T^f = \tilde{F} - \tilde{F}|_T = \int_{T^{-1}i\infty}^{i\infty} \omega_f = 0 \quad (\text{whenever the integral converges}), \quad (4)$$

the contour collapsing because T *fixes* the cusp $i\infty$. This vanishing is the precise content of “no monodromy around the cusp”: $C_T^f = 0$ says $\tilde{F}$ is invariant under the *full* T -action, i.e. a *flat section of the local system V_n around the cusp*. It is emphatically *not* the bare periodicity $\tilde{F}(\tau) = \tilde{F}(\tau + 1)$ — that statement is false, since $\tilde{F}(\tau) - \tilde{F}(\tau + 1) = \int_\tau^{\tau+1} \omega_f \neq 0$. The monodromy of $\tilde{F}$ around the cusp lives in V_n (it is the unipotent T acting on X, Y), not in the scalar coordinate $q = e^{2\pi i\tau}$; conflating the two is the one trap here.

Two representatives of one class — and which one §5–§6 use. Equation (4) is the *cuspidally anchored* value, and it vanishes identically for every branch that converges. So the live content of “solve $C_T = 0$ ” below is not carried by this representative at all, but by a *cohomological* one: the finite-basepoint cocycle $C_T^f(\tau_0) = \int_{\tau_0-1}^{\tau_0} \omega_f$ (the form used in §6), whose X^n -component is the constant Fourier coefficient $c_f(0)$. The two differ by the coboundary that moves the basepoint off the cusp. “Re-impose $C_T = 0$ ” then means: correct the available finite-basepoint branch back to a flat section. For holomorphic f this is free, inherited from the convergent cusp anchor (§5). For weakly holomorphic f , the *only* thing that diverges is the *classical* cusp-anchored primitive $\int^{i\infty} \omega_f$ (its contour runs into the cusp, where $f \sim q^{-1}$); the cusp-regular branch still *exists* and is finite whenever $c_f(0) = 0$ (§6), but it can no longer be read off by integrating from $i\infty$ — so it must instead be *solved for* on finite contours, the solution being the T -segment integral. Nothing in the construction itself diverges: every contour in play runs between finite points of $\mathbb{H}$ (where f is holomorphic), which is the whole “manifestly finite” point. What diverges is only the classical object one is generalizing away from.

This is distinguished, not arbitrary — but it is worth being exact about *in what sense*, because “canonical” and “forced” are not the same claim and it is easy to smuggle one in under the other. Three levels, from most to least compulsory.

1. **Forced (you must pick *something*).** The class $[C^f]$ is all you have basepoint-independently, and a class contains no polynomial. To get numbers at all you must descend to a representative, i.e. fix a branch of $\tilde{F}$. This step is not optional: no representative, no period polynomial.
2. **Forced (it must be T , not S).** This is the real compulsion, and it is structural, not aesthetic: *T is essentially the only generator whose value you are permitted to normalize away.* Normalizing the g -value means solving $\delta_g P := P|_g - P = C_g$ for $P \in V_n$. For $g = T$ this is almost always solvable, because T acts unipotently and δ_T is nilpotent, hence near-surjective — it fails only on a one-dimensional obstruction (the cusp residue; §6). For $g = S$ it is almost always *unsolvable*: S acts as an involution, so $\text{im } \delta_S = \ker(1 + S)$ is a fixed proper subspace, and a generic C_S generically does not lie in it. (For the actual period cocycle the S -obstruction *does* vanish — that is the S -relation, §7 — so one could sometimes solve $\delta_S P = C_S$; the point is that doing so would trivialize the very S -value that carries the periods, whereas δ_T removes the cusp monodromy while leaving the S -value with content. T is the direction one can spend *usefully*.) So among the generators, T is not the pretty choice — it is the one whose normalization both exists in general and preserves the period content, and the surviving S -value carries that content precisely because it is the one we do not spend.
3. **Canonical given the goal (which T -slice).** Even granting “normalize via T ,” the condition still has slack: $\delta_T P = C_T$ determines P only up to $\ker \delta_T = \mathbb{C}Y^n$ (§6), so $C_T = 0$ is an affine *family* of representatives, not one. That T is the generator to spend is settled by level 2; *why exactly-zero, and not some other T -slice*, is settled by the external definition of r_f , and is important enough to get its own section (§4). The short version: $C_T = 0$ is exactly the family of branches agreeing with the cusp-anchored Eichler integral, and any other normalizable slice shifts the *interior* coefficients of the S -value — the very critical L -values — by a spurious offset.

The honest one-sentence status: the truly invariant object is the parabolic class in $W_{k-2}/\mathbb{C}(X^n - Y^n)$ (equivalently $C_T = 0$ up to $\mathbb{C}Y^n$); “ $C_T = 0$ exactly” is the convenient concrete lift of it, forced *relative to the goal* of continuing the classical r_f , and forced *structurally* in that T is the only direction one can normalize at all.

In one line. $C_T = 0 \iff$ the primitive is single-valued around the cusp. Two things make this the choice: structurally, T is the only generator whose value can be normalized away (S cannot); and relative to the goal, no-cusp-monodromy is exactly the branch whose S -obstruction is the classical period polynomial. The leftover $\mathbb{C}Y^n$ slack is the one honestly conventional bit.

Once that branch is fixed, the surviving datum $C_S = \tilde{F} - \tilde{F}|_S$ measures non-modularity under $\tau \mapsto -1/\tau$, and that S -obstruction of the cusp-regular primitive is the period polynomial. This is the content of (1): the endpoints 0 and $i\infty$ are an S -pair, and $r_f = \tilde{F}|_S - \tilde{F}$ read at the cusp-anchored branch.

4 Why *exactly* zero: the argument that $C_T = 0$ reproduces r_f

This is the main argument, and it deserves to be spelled out rather than asserted, because it is the only thing that makes $C_T = 0$ the *right* normalization rather than merely *a* normalization. The claim is goal-relative and that is not a weakness; it is the honest shape of the statement.

Step 1: the classical object is one branch's S -value. Write $\tilde{F}_\infty$ for the cusp-anchored primitive — the Eichler integral (2), whose fixed (upper) endpoint is the cusp $i\infty$ and whose free lower limit is the running variable τ . By definition (1),

$$r_f = \int_0^{i\infty} \omega_f = \tilde{F}_\infty|_S - \tilde{F}_\infty = -C_S^f \text{ evaluated at the branch } \tilde{F}_\infty.$$

So “the period polynomial” is not the S -value of *some* representative; it is the S -value of *this particular* branch. Any argument that $C_T = 0$ gives r_f must show that the $C_T = 0$ condition selects the branch $\tilde{F}_\infty$ — or selects it closely enough.

Step 2: $\tilde{F}_\infty$ satisfies $C_T = 0$, but $C_T = 0$ does not pin it uniquely. For holomorphic f , $\tilde{F}_\infty$ is T -invariant in the full slash, so $C_T^f = \tilde{F}_\infty - \tilde{F}_\infty|_T = 0$: the classical branch is *a* solution of the normalization. It is not the only one. Two branches with the same T -value differ by R with $\delta_T R = 0$, i.e. $R \in \ker \delta_T = \mathbb{C}Y^n$. So “ $C_T = 0$ ” carves out a *one-parameter family* of branches, $\tilde{F}_\infty + \mathbb{C}Y^n$, and $\tilde{F}_\infty$ is one point of it.

Step 3: the residual ambiguity is exactly the ambiguity of r_f itself. Moving along the family, $R = cY^n$, changes the S -value by

$$R|_S - R = c(Y^n|_S - Y^n) = c(X^n - Y^n) \quad (Y|_S = X).$$

This is the punchline of the whole matter: the freedom that $C_T = 0$ fails to remove acts on the S -value *only* through $\mathbb{C}(X^n - Y^n)$, i.e. only on the two boundary monomials. Therefore

$C_T = 0 \text{ reproduces } r_f \text{ exactly as an element of } W_{k-2}/\mathbb{C}(X^n - Y^n).$

That is not “ r_f up to slop”; it is r_f *at the precision at which r_f is itself well-defined*, since the classical period polynomial carries the same intrinsic $\mathbb{C}(X^n - Y^n)$ ambiguity (the even part is only defined modulo the period polynomial of the Eisenstein series). The interior coefficients $\ell \in \{1, \dots, n-1\}$ — the critical L -values — are pinned exactly.

Step 4: no genuinely different slice does this. Suppose one normalized along a different rule, landing on a branch $\tilde{F}_\infty + R$ with $R \notin \mathbb{C}Y^n$. The resulting S -value is $r_f - (R|_S - R)$, and $R|_S - R \in \mathbb{C}(X^n - Y^n)$ forces $R \in \mathbb{C}Y^n + \ker(\delta(\cdot)(S))$. Unwinding, the only way to keep the interior coefficients equal to the classical critical L -values is to have chosen, up to $\mathbb{C}Y^n$, the cusp-anchored branch after all. Any *other* normalizable slice shifts at least one interior coefficient by a nonzero offset sourced purely by the choice of branch: you obtain a clean polynomial satisfying clean relations, whose interior coordinates are the true critical L -values *plus a spurious constant*. Clean, well-defined — and not the period polynomial.

What is and isn't being claimed. The argument is relative to the target: *given* that we are continuing the classical r_f , $C_T = 0$ is the unique normalization (mod the intrinsic $\mathbb{C}(X^n - Y^n)$) that does so. It is *not* a claim that $C_T = 0$ is forced in a vacuum — absent any attachment to the classical object, another slice would define an equally consistent “period polynomial.” The cusp is privileged not by pure algebra but because the object being generalized was built at the cusp. This is also why, for weakly holomorphic f , one cannot simply “use $\tilde{F}_\infty$ ”: it diverges (§5), so the branch that would have agreed with the cusp-anchored one is no longer obtainable by integrating from $i\infty$. Solving $\delta_T P = C_T^f$ is precisely the algebraic continuation of “be the cusp-regular branch” into the regime where the cusp anchor is gone — and that solved-for P is the T -segment integral.

5 Why this is invisible classically but a step here

For a *holomorphic* cusp form, the cusp-anchored primitive (2) is automatically a flat section around the cusp: $f(\tau + 1) = f(\tau)$ and $\tilde{f}$ decays at $i\infty$, so the integral defining $\tilde{F}$ converges and the substitution $z \mapsto z - 1$ gives $\tilde{F}|_T = \tilde{F}$ (the shift $\tau \mapsto \tau + 1$ exactly compensates the polynomial shear $X \mapsto X + Y$). Hence

$$C_T^f = 0 \quad \text{for free, with no choice made.}$$

Nobody in the classical theory ever “normalizes” anything; they simply write $\int_0^{i\infty}$, which silently bases the primitive at the cusp, which silently enforces $C_T = 0$. The normalization was always present — it was just inherited from the geometry rather than imposed.

For a *weakly holomorphic* form like $\hat{\Delta} = q^{-1} + O(q^2)$, the q^{-1} term makes (2) diverge at $i\infty$: there is no convergent cusp-anchored primitive to inherit periodicity from. The naive primitive you *can* write down acquires a genuine winding at the cusp, so $C_T^f \neq 0$ as handed to you. The move of the construction is then forced and concrete:

$$\text{solve } \delta_T P := P|_T - P = C_T^f \quad \text{for } P \in V_n, \quad \text{then subtract } P. \quad (5)$$

Subtracting P re-installs periodicity by hand — it manufactures the cusp-regular branch algebraically, since the analytic one no longer exists. **The polynomial P you solve for is exactly the T -segment integral** (the second integral in the main theorem, with its Bernoulli/Hurwitz kernel). That integral is not a regulator; it is the correction that turns the available primitive into the cusp-regular one.

In one line. *Classically, “base at $i\infty$ ” enforces $C_T = 0$ silently. When the form is weakly holomorphic the cusp is no longer a place you can stand, so the same normalization must be solved for — and the solution is the T -segment integral.*

6 The leftover constant, and the obstruction

Two loose ends, both meaningful.

The residual freedom is the constant of integration. Equation (5) does not determine P uniquely: δ_T has a one-dimensional kernel,

$$\ker \delta_T = \{P : P|_T = P\} = \mathbb{C} Y^n$$

(the only T -invariant polynomials are powers of Y , since $Y|_T = Y$ while $X|_T = X + Y$). So P is fixed only modulo $\mathbb{C} Y^n$, and correspondingly $r_f = C_S - (P|_S - P)$ is well-defined only modulo

$$\delta(\cdot)(S) [\mathbb{C} Y^n] = \mathbb{C} (Y^n|_S - Y^n) = \mathbb{C} (X^n - Y^n) \quad (\text{since } Y|_S = X).$$

This is not noise to be discarded: it is the genuine *constant of integration* of the primitive, the one datum a single antiderivative cannot pin. It touches only the boundary monomials X^n, Y^n , which is exactly why the interior coefficients ($\ell \in \{1, \dots, n-1\}$) are unambiguous and the main theorem ranges over them. (This is the point that the word “gauge” actively *hides*: it is data, not redundancy.) The moral, restated from §3: the genuinely invariant object extracted from $[C^f]$ is not a polynomial but the parabolic class

$$[r_f] \in W_{k-2}/\mathbb{C} (X^n - Y^n) \quad (\cong H_{\text{par}}^1(\text{SL}_2(\mathbb{Z}); V_n)),$$

and “ $C_T = 0$ exactly, with a chosen $\mathbb{C} Y^n$ representative” is one concrete lift of that class to an honest polynomial. The forced content lives in the quotient; the lift is the convenient part.

The normalization can fail to exist — and that failure is correct. Equation (5) is solvable iff C_T^f lies in $\text{im } \delta_T$. The cokernel is one-dimensional,

$$V_n / \text{im } \delta_T \cong \mathbb{C} \cdot [X^n],$$

and the component of C_T^f there is the constant Fourier coefficient $c_f(0)$ — the residue of ω_f at the cusp. So:

a cusp-regular primitive exists $\iff c_f(0) = 0 \iff f \in S_k^!$ (cuspidal, incl. weakly holomorphic).

When $c_f(0) \neq 0$ (the Eisenstein case), *no* winding-free branch exists, no $C_T = 0$ normalization exists, and the construction correctly declines to produce a period polynomial. The normalization and its obstruction are the two ends of a single fact: you can remove the cusp monodromy precisely when there is no residue forcing it.

7 What survives once $C_T = 0$ (the normal form)

A bonus, and the part of the original “gauge” instinct worth keeping. With $C_T = 0$ the cocycle is determined by the single polynomial $r := C_S$, because $\text{PSL}_2(\mathbb{Z}) = \langle S \rangle * \langle ST \rangle$ is a free product of the two torsion groups and the cocycle relation telescopes. Iterating $C_{g_1 \dots g_m} = \sum_i C_{g_i}|_{g_{i+1} \dots g_m}$ and killing every T -letter, a general element $\gamma = S T^{n_1} S T^{n_2} \dots S T^{n_L}$ has

$$C_\gamma^f = \sum_{j=1}^L r|_{h_j}, \quad h_j = T^{n_j} S T^{n_{j+1}} \dots S T^{n_L}. \quad (6)$$

The value on a word of S -length L is a finite sum of L transported copies of the one polynomial r — not two; one per S -syllable. The two group relations prune *short* words: $S^2 = 1$ gives $r|_S = -r$, and $(ST)^3 = 1$ gives $r|(1 + TS + (TS)^2) = 0$; these are the classical period relations and they cut redundancy but do not shrink a generic length- L sum below $\sim L$ terms.

Caveat for the meromorphic case. The collapse in (6) uses $C_{S^2} = 0$ and path-independence, both statements about a fixed homotopy class of contour. With interior poles the free-product normal form still holds algebraically, but “the value of γ ” becomes per-chamber; crossing a pole adds a residue jump on top of the telescoped sum. The normal form is the within-chamber skeleton, not the whole answer.

8 Status of the claims: what is, and isn’t, a theorem

It is worth asking your own question directly — *if $C_T = 0$ is really needed, shouldn’t that be a proposition?* The useful answer is that the demand forces a sorting, because several statements of different strength have been circling, and only some are theorems. One of them is a non-claim; two are genuine propositions provable by linear algebra and cohomology with *no* special functions; one is the analytic theorem of the paper, and that last is the only one whose proof is “the incomplete gammas match.” Keeping these apart is the whole point: a proof attached to the wrong claim is a seam.

(A) “ $C_T = 0$ is necessary” — *not* a theorem. There is no absolute necessity to prove, and attempting it is the trap. As §3–§4 show, the necessity is goal-relative: absent any attachment to the classical r_f , a different normalizable slice defines a different, internally consistent polynomial obeying clean relations. So the statement one is tempted to elevate is precisely the one that cannot be a theorem. State it as a non-claim and move on.

(B) Existence and uniqueness of the normalization — a proposition (linear algebra).

Let f be weakly holomorphic of weight k , $n = k - 2$, with $c_f(0) = 0$. Then there exists $P \in V_n$ with $\delta_T P = C_T^f$, unique modulo $\ker \delta_T = \mathbb{C}Y^n$. Consequently the cocycle $C^f - \delta P$ has vanishing T -component, and its S -value is well-defined in $W_{k-2}/\mathbb{C}(X^n - Y^n)$. If $c_f(0) \neq 0$, no such P exists.

Proof. Pure linear algebra on the unipotent operator $\delta_T = (\cdot)|_T - \text{id}$. In the basis $\{X^{n-j}Y^j\}$, δ_T is strictly upper-triangular (it lowers the X -degree), so $\text{im } \delta_T = \langle X^{n-j}Y^j : 1 \leq j \leq n \rangle = \{Q : [Q]_{X^n} = 0\}$ has codimension one with $\text{coker} \cong \mathbb{C}[X^n]$, and $\ker \delta_T = \mathbb{C}Y^n$ has dimension one. Solvability is the vanishing of the coker-component of C_T^f , which Lemma (residue) identifies with $c_f(0)$; uniqueness mod $\mathbb{C}Y^n$ is $\ker \delta_T$. The induced ambiguity on the S -value is $\delta(\cdot)(S)[\mathbb{C}Y^n] = \mathbb{C}(Y^n|_S - Y^n) = \mathbb{C}(X^n - Y^n)$. $\square$

(C) The normalization computes the period polynomial — a proposition (cohomology, no special functions).

For $f \in S_k$ holomorphic, the S -value of the $C_T = 0$ representative equals the classical period polynomial $r_f = \int_0^{i\infty} \omega_f$ as an element of $W_{k-2}/\mathbb{C}(X^n - Y^n)$.

Proof. This is §4, Steps 1–3, boxed. The cusp-anchored branch $\tilde{F}_\infty$ converges (holomorphic f) and is periodic, so it satisfies $C_T = 0$ and its S -value is exactly r_f by (1). Any other $C_T = 0$ branch differs from $\tilde{F}_\infty$ by an element of $\ker \delta_T = \mathbb{C}Y^n$ (B), which shifts the S -value only within $\mathbb{C}(X^n - Y^n)$. Hence all $C_T = 0$ representatives agree with r_f in the quotient. $\square$

Note the proof uses no q -expansion, no Mellin transform, and no incomplete gamma function. C is true at the level of the cocycle, before any analytic object is computed; it is what makes the construction *the* period polynomial rather than merely *a* polynomial.

(D) The analytic agreement — the paper’s theorem, and the only place the incomplete gammas enter.

The $C_T = 0$ L -integral, read at integer $s = \ell + 1$, equals BFK’s completed L -function: expanding the surviving T -segment integral yields the incomplete-gamma sums $\sum_{n \neq 0} c(n) [\Gamma(\ell, 2\pi n)/(2\pi n)^\ell + i^k \Gamma(k - \ell, 2\pi n)/(2\pi n)^{k-\ell}]$.

Proof sketch. This is where “the incomplete gammas match”: the Hurwitz-zeta kernel of the T -segment unfolds, term by term in the q -expansion, into upper incomplete gammas $\Gamma(\ell, 2\pi n)$, reproducing the BFK completed L -function. This is the paper’s Theorem (**thm:bfk-eq**), proved by the Hurwitz-unfolding lemma.

Why D is not B or C. D presupposes C. C says the S -value *is* the period polynomial (a statement about a polynomial and its class); D takes the coefficients of that polynomial — which are critical L -values — and writes *those* as incomplete-gamma sums. The chain is $B \Rightarrow C \Rightarrow D$: existence of the normalized representative, identification of its S -value as the period polynomial, expansion of that polynomial’s coefficients into the analytic completed L -function. The incomplete gammas live only on the last arrow. Attaching them to C (or to “why $C_T = 0$ ”) would be proving the wrong statement.

To verify before any of this is a Proposition in the *paper*. C as stated compares against a pre-existing classical r_f , which exists only for $f \in S_k$. For weakly holomorphic $f \in S_k^! \setminus S_k$ there is no prior period polynomial to agree with, so C must be *replaced* by an extension-uniqueness statement:

(C', to verify) The $C_T = 0$ S -value is the unique linear extension of $r_{(\cdot)}$ from S_k to $S_k^!$ that is Hecke-equivariant [/computed by the T -segment integral / satisfies the period relations mod $\mathbb{C}(X^n - Y^n)$ — pick the characterization you can actually prove].

This is subtler than B or holomorphic-C and is where a referee will push. It should be checked against the verification numerics (the $k = 12$ $\Delta, \hat{\Delta}$ data) before the word “Proposition” goes near it.

The whole note in five sentences. The period cocycle is the coboundary of the Eichler integral $\tilde{F}$, a primitive of ω_f ; choosing a representative means choosing a branch of $\tilde{F}$, and you are forced to choose *something* because a class holds no polynomial. Among the generators only T can be normalized away at all — δ_T is near-surjective while δ_S is not — so the surviving S -value carrying the period content is forced structurally, not by taste. The condition $C_T = 0$ says $\tilde{F}$ is single-valued around the cusp, and that branch is the one whose S -obstruction reproduces the classical r_f , so it is forced *relative to the goal* of continuing r_f to weak forms — though the last

$\mathbb{C}Y^n$ of slack is honestly conventional. For holomorphic cusp forms this branch is inherited free by basing the integral at $i\infty$; for weakly holomorphic forms the cusp anchor diverges, so the cusp-regular branch is re-installed by solving $\delta_T P = C_T$, whose solution is the T -segment integral. The genuinely invariant object is the parabolic class $[r_f] \in W_{k-2}/\mathbb{C}(X^n - Y^n)$, the whole construction exists iff the cusp residue $c_f(0)$ vanishes, and “ $C_T = 0$ exactly” is a convenient lift of that class to a polynomial.